

LATENT FIXTURE REASONING

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ABSTRACT

Assembly robots often treat support relations and fixtures as visible scene geometry or as constraints supplied before planning. This paper studies the missing case: a part fails to move because an unobserved bolt, latch, adhesive patch, or hidden stop is still constraining it. We propose *latent fixture reasoning*: infer a discrete hidden fixture cause from small manipulation outcome residuals, then choose the release or reorientation action implied by that cause. A bounded literature sweep over 1168 arXiv/Crossref records suggests that the nearest work covers contact-rich manipulation, constraint reasoning, tactile inference, active perception, and fixture-aware assembly, but rarely treats hidden fixtures as an explicit action-selecting state variable. In a deterministic toy benchmark, geometry-only manipulation collides with many fixture modes, while latent fixture reasoning sharply reduces wrong release actions. The evidence is modest, but it exposes a useful state abstraction for robots that must learn what is holding the world in place by acting on it.

1 INTRODUCTION

When a robot pulls on a part and it does not move, the failure is not just a low-level control error. It can be evidence that something unseen is holding the part in place. In assembly, repair, household manipulation, and field robotics, that hidden cause may be a bolt, a latch, adhesive, a cable, a clamp, or a geometric stop outside the camera view. A planner that represents only visible geometry can repeatedly choose unsafe or unproductive actions.

We call this problem *latent fixture reasoning*. The key object is an interpretable hidden fixture variable. It is latent because the robot does not observe the fixture directly. It is a fixture because it constrains motion. It is reasoning, rather than only prediction, because the inferred cause changes the next action: unbolt, unlatch, heat, reorient, or pull.

The paper makes three contributions. First, it separates latent fixture causes from generic uncertainty in manipulation. Second, it defines a small posterior interface from probe residuals to fixture hypotheses. Third, it provides a minimal deterministic simulation showing why this state variable can reduce wrong fixture actions.

2 PROBLEM

Let x denote the visible object state and let z be an unobserved fixture mode in $\{\text{free, bolt, latch, adhesive, stop}\}$. A robot chooses a probe action a_p and observes an outcome residual o : displacement, lift compliance, twist compliance, and thermal softening response. The robot then chooses a task action a_t from direct pull, unbolt, unlatch, heat-and-peel, or reorient. The central question is not whether the robot can predict the next state in general. It is whether o identifies a fixture cause well enough to avoid the wrong physical release action.

3 MECHANISM

The proposed controller maintains a small posterior over fixture modes,

$$p(z \mid o) \propto p(o \mid z)p(z).$$

The likelihood model can be hand-written, learned, or calibrated from robot data. In this paper we use a hand-written model only to isolate the mechanism. The controller is allowed to perform a reversible probe before committing to a release action. If confidence is low, it performs an additional reversible probe rather than applying a high-force release action.

This differs from a generic world model in two ways. First, the latent variable has a robotics meaning: it is a hidden support or fixture cause. Second, the variable is directly coupled to an action menu. A high posterior on bolt selects unbolt; a high posterior on adhesive selects heat-and-peel; a high posterior on hidden stop selects reorientation.

4 EVIDENCE

We implemented a deterministic toy benchmark with 2,000 seeds. Each seed samples one fixture mode, generates noisy probe residuals, and evaluates three policies. The geometry-only policy assumes visible geometry is complete. The reactive policy first fails, then uses a single heuristic residual. The latent fixture policy infers the fixture before choosing a release action.

Policy	Success	Coll.	Wrong	Steps	Return	Conf.
Geometry only	621	1379	1379	3.689	-0.696	0.000
Reactive	1612	1767	388	5.758	0.332	0.207
Latent fixture	1995	5	5	7.077	0.871	0.941

Table 1: Toy hidden-fixture benchmark. The mechanism is valuable when a visible-geometry policy cannot distinguish free motion from a hidden fixture constraint.

The absolute numbers should not be read as benchmark performance. They show a mechanism: when the hidden cause selects a different release action, estimating that cause before committing can reduce physical mistakes.

5 RELATED WORK BOUNDARY

The literature sweep covers contact-rich manipulation, task-and-motion planning, tactile inference, support-relation reasoning, robot world models, and assembly fixture planning. These areas provide the hostile prior. The narrow boundary for this paper is an interpretable latent fixture graph whose nodes are not merely unobserved state, but action-selecting causes of constraint.

6 LIMITATIONS

The experiment is a toy model with hand-coded fixture types and synthetic residuals. It does not prove that the abstraction scales to real assemblies. The next required step is a physical or high-fidelity simulated benchmark where fixtures are hidden from perception but revealed through controlled probes.

7 CONCLUSION

Latent fixture reasoning asks robots to treat failed motion as evidence about what hidden support is still constraining the world. The contribution is a small but useful shift in representation: from visible geometry plus generic uncertainty to an interpretable hidden fixture cause that chooses the next safe manipulation action.

REFERENCES

The accompanying `docs/related_work_matrix.csv` contains the bounded recovery literature sweep used for this mechanism note.